

1 Preface

From today(Nov 11) i will begin to re-learn quantum mechanics and start my note working ,try to fix the whole quantum world in my head.

2 Wavefunction

2.1 Begin of the probability wave

Discribe the probability of a particle that will appear in a place.And from the principle of quantum mechanics , consider a situation that don't producing any or annihilation any particle.

$$\int \phi^* \phi \, \varepsilon = 1 \quad (1)$$

That means the whole probability of a particle is 1 for every possible position. Here are some equations will be used in the next learning:

$$(\overline{\Delta u})^2 = \overline{(u^2)} - (\bar{u})^2 \quad (2)$$

$$\overline{\Delta u} = \sqrt{\overline{u^2} - \bar{u}^2} \quad (3)$$

These are the simplest math equations but will be used in **uncertainty relation**.

2.2 wave function

we call the wave function which is differ in a constant multiplied are the same function:

$$\phi = C\psi \quad (4)$$

so we call the principle is **the probability of the function waves are relatively**.

2.3 momentum

We all know how to define the momentum of a particle , but (consider it is the relearn note) we know the uncertainty principle,so by using **Fourier transformation**:

$$f(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(x) e^{-ikx} dx \quad (5)$$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(k) e^{ikx} dk \quad (6)$$

and define the momentum:

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int \phi(x) e^{-\frac{ipx}{\hbar}} dx \quad (7)$$

What do $\phi(p)$ express? It expresses the **format of wave function in the space of momentum**. So we can calculate the momentum by:

$$\bar{p} = \int_p \phi(p)^* p \phi(p) \quad \epsilon \quad (8)$$

This integral is defined on the space of momentum, the reason of why we not to choose to calculate the average momentum by using $\int \phi(x)^* p \phi(x)$ is the momentum can not be certain with x at the same time.

And we also consider about the de Broglie relation:

$$p = \frac{h}{\lambda} \quad (9)$$

2.4 uncertainty relation

The strict define of uncertainty relation:

$$\Delta A \Delta B \geq \frac{1}{2} |\overline{[A, B]}| \quad (10)$$

The $\Delta A \Delta B$ here are the $\sqrt{(\Delta A)^2 \times (\Delta B)^2}$.

2.5 Average of Mechanic Variable

The define is :

$$\int \phi^* A \phi = \bar{A} \quad (11)$$

2.6 coordinate representation

How to express coordinate representation? It just like the relationship between physics object and coordinate base. Like how to get the part of a tensor or a part of a vector. The $\phi(\mathbf{t})$ is the physics object and $\phi(\mathbf{x})$ is this object in x space, and $\phi(\mathbf{p})$ is the object in momentum space.

And the base of these projection are the base in Hilbert space.

Hilbert space: the every possible state of wave function ϕ .

CSCO: a set of commuting operator, and the co-eigenstate is a base of Hilbert space.

3 Shrödinger equation

$$i\hbar \frac{\partial \phi(r, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(r, t) \right] \phi(r, t) \quad (12)$$

3.1 propagator

Because Shrödinger equation only have the 1 class derivative of time t so if we set the wave function **of free particle** at $t = 0$: $\phi(r, 0)$ we can get we any time wave function $\phi(r, t)$:

$$\phi(r, t) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} \int \phi(p) e^{\frac{ipr - iEt}{\hbar}} d^3p \quad E = \frac{p^2}{2m} \quad (13)$$

$$\phi(r, 0) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} \int \phi(p) e^{\frac{ipr}{\hbar}} d^3p \quad (14)$$

reverse tranformation: $\phi(p) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} \int \phi(r, 0) e^{\frac{-ipr}{\hbar}} d^3r$ put back to (13) (15)

$$\phi(r, t) = \frac{1}{2\pi\hbar^3} \int d^3p \int d^3r' \phi(r', 0) e^{\frac{ip(r-r') - iEt}{\hbar}} \quad (16)$$

The reason of from eq(14) to eq(15) is **Fourier transformation don't rely on time t** and for any common situation:

$$\psi(r, t) = \frac{1}{2\pi\hbar^3} \int d^3r' \int d^3p \phi(r', t') e^{\frac{ip(r-r') - iE(t-t')}{\hbar}} \quad (17)$$

rewrite it into form :

$$\psi(r, t) = \int d^3r' G(r, r', t, t') \phi(r', t') \quad (18)$$

$$G = \frac{1}{2\pi\hbar^3} \int e^{\frac{ip(r-r') - iE(t-t')}{\hbar}} d^3p \quad (19)$$

and the G in there is called propagator. For an eigenstate of $\hat{H}$ the wave function can be write as: $\psi(r, t) = \frac{1}{2\pi\hbar^{\frac{3}{2}}} e^{\frac{ipr - iEt}{\hbar}}$.

Now look back at eq(18) we can express this equation as :Wave function of time at t is all wave function of the particle in time t' and r' coherent superposition then spreaded to r .

3.2 Time Factor

We set $\psi(r, t) = \psi(r)f(t)$ and consider a simple situation : V in $\hat{H}$ don't explicit t . And $\psi(r, t)$ is the eigenstate of E .

$$i\hbar\psi(r)\frac{\partial f(t)}{\partial t} = [-\frac{\hbar^2}{2m}\nabla + V]\psi(r, t) \quad (20)$$

$$Ef(r, t) = [-\frac{\hbar^2}{2m}\nabla + V]\psi(r, t) \quad (21)$$

$$f(t) = e^{\frac{-iEt}{\hbar}} \quad (22)$$

so the $\psi(r, t)$ can be rewirte as $:\psi_E(r)f(t)$ For a more normalized situation, we can write any $\psi(r, t)$ into eigenstate of E: $\psi_{E_n}(r)$:

$$\psi(r, t) = \sum_n C_n \psi_{E_n} f_n(t) = \sum_n C_n \psi_{E_n} e^{-\frac{i E_n t}{\hbar}} \quad (23)$$

$$|C_n|^2 = \int \psi_{E_n}^* \psi(r, 0) \quad (24)$$

3.3 Stationary state

We call a wave function in form $\psi(r, t) = \psi(r) e^{-i \frac{E t}{\hbar}}$ as the stationary state wave function. When a particle stay in stationary state the energy of this particle is exactly can be determined. And the particle have these nature satisfied when it is a stationary particle:

(1) the probability density is not go with time:

$$\psi^* \psi = \rho(r) \quad (25)$$

$$(26)$$

(2) the probability flux is not go with time :

$$j_n(r, t) = \frac{i\hbar}{2m} [\psi_n \nabla \psi_n^* - \psi_n^* \nabla \psi_n] = j_n(r) \quad (27)$$

(3) any average of dynamical variables which don't dependent on time will not dependent on time:

$$\bar{F} = \int \psi(r, t)^* F \psi(r, t) = \int \psi(r)^* F \psi(r) \quad (28)$$

(4) any observed value from observable dynamical variables don't explicit t will not dependent on time.

4 Operator

4.1 The Operator Algebra

Operators normally don't satisfy the law of multiple exchange. So we define the commutator:

$$[A, B] = AB - BA \quad (29)$$

And also can de defined the linearity of operators:

$$(c_1 A + c_2 B) \psi = c_1 A \psi + c_2 B \psi \quad (30)$$

4.2 commutator

Commutator is one of the important concept in QM, and there are some qualities commutator have:

$$[x, y(p(x))] = i\hbar \frac{\partial y}{\partial p(x)} \quad (31)$$

$$[p(x), y(x)] = -i\hbar \frac{\partial y}{\partial x} \quad (32)$$

The equation(31) is the equation in coordinate representation p, and equation(32) is in the coordinate representation x. So we can find the basic law:

$$[A(x), B(A(x))] = i\hbar \frac{\partial B}{\partial A(x)} \quad (33)$$

and from the upper discussion we know there is a strong relationship between commutator and uncertainty relation at equation(10).

And the function of a variable will commuted with another function of the same variable

$$[A(f), B(f)] = 0 \quad (34)$$

4.3 Some unique transform in QM

Most of mechanics operators is hermited, $\hat{A}^\dagger = \hat{A}$ for define

$$\int \phi^* \hat{A}^\dagger \psi = \int (\hat{A} \psi)^* \psi = \int \psi^* \hat{A} \psi \quad (35)$$

4.4 angular momentum in QM

The operator of angular momentum: $\hat{l} = r \times \hat{p} = r^i \hat{p}^j \varepsilon_{ijk}$ Rewrite them in a sphere coordinate system, noticed that the form of l_z is simple and **we know now, the eigenstate of l_z is a Spherical harmonic function Y_{lm} . NOTICE**: the eigentstate of l_z can be a part of Spherical harmonic function: $\psi = \frac{1}{\sqrt{2\pi}} e^{im\phi}$, and the eigenstate of l^2 can be another part of Spherical harmonic function. **So Spherical harmonic function is the co-eigenstate of both operator: $\hat{l}_z, \hat{l}^2$.**

$$\hat{l}_z Y_{lm} = m\hbar Y_{lm} \quad (36)$$

$$\hat{l}^2 Y_{lm} = l(l+1)\hbar^2 Y_{lm} \quad (37)$$

$$[\hat{l}^2, \hat{l}_i] = 0 \quad (38)$$

$$[l_i, x^j] = i\hbar \varepsilon_{ijk} x^k \quad (39)$$

$$[\hat{l}_i, \hat{l}_j] = i\hbar \varepsilon_{ijk} \hat{l}_k \quad (40)$$

And the define of Shperiacl harmonic function:

$$Y_{lm}(\theta, \phi) = (-1)^m \sqrt{\frac{(l-m)!(2l+1)}{4\pi(l+m)!}} P_l^m(\cos\theta) e^{im\phi} \quad (41)$$

$$l = 0, 1, 2, \dots, m = -l, -l+1, \dots, l-1, l.$$

And the Orthogonal normalization condition is written as:

$$\int Y_{lm}^* Y_{l'm'} \epsilon = \delta_{ll'} \delta_{mm'} \quad (42)$$

Now I will list a important example :

$$\begin{aligned} \hat{l}^2 - \hat{l}_z^2 &= \hat{l}_x^2 + \hat{l}_y^2 \text{ and consider a situation } (\psi = Y_{lm}): \int \psi^* \hat{l}_x^2 \psi \\ &= \int \psi^* \frac{1}{i\hbar} [\hat{l}_y, \hat{l}_z] \hat{l}_x \psi = \frac{1}{i\hbar} \int \psi^* (\hat{l}_y \hat{l}_z - \hat{l}_z \hat{l}_y) \hat{l}_x \psi \\ &= \frac{1}{i\hbar} \int \psi^* \hat{l}_y \hat{l}_z \hat{l}_x \psi - \phi^* \hat{l}_z \hat{l}_y \hat{l}_x \psi \\ &\text{consider } \hat{l}_z \hat{l}_x = \hat{l}_x \hat{l}_z + i\hbar \hat{l}_y \\ &= \frac{1}{i\hbar} \int \psi^* \hat{l}_y (\hat{l}_x \hat{l}_z + i\hbar \hat{l}_y) \psi - \psi^* \hat{l}_z \hat{l}_y \hat{l}_x \psi \\ &= \frac{1}{i\hbar} \int \psi^* \hat{l}_y \hat{l}_x \hat{l}_z \psi + i\hbar \psi^* \hat{l}_y^2 \psi - \psi^* \hat{l}_z \hat{l}_y \hat{l}_x \psi \\ &= \frac{1}{i\hbar} \int \psi^* \hat{l}_y \hat{l}_x m\hbar \psi + i\hbar \psi^* \hat{l}_y^2 \psi - m\hbar \psi^* \hat{l}_y \hat{l}_x \psi \\ &= \int \psi^* \hat{l}_y^2 \psi \\ &\text{so we get} \end{aligned}$$

$$\overline{\hat{l}_x^2} = \overline{\hat{l}_y^2} \quad (43)$$

Now consider the first equation :

$$\begin{aligned} \overline{\hat{l}^2} + \overline{\hat{l}_z^2} &= 2\overline{\hat{l}_y^2} = 2\overline{\hat{l}_x^2} \\ \text{and } \int \psi^* \hat{l}^2 \psi &= \int \psi^* l(l+1)\hbar^2 \psi = l(l+1)\hbar^2 ; \int \psi^* \hat{l}_z^2 \psi = \int (\hat{l}_z \psi)^* \hat{l}_z \psi = m^2 \hbar^2 \\ \text{so} \end{aligned}$$

$$\overline{\hat{l}_x^2} = \overline{\hat{l}_y^2} = \frac{1}{2}(l(l+1) - m^2)\hbar^2 \quad (44)$$

We know the average of angular momentum $\hat{l}_x$ and $\hat{l}_y$ is : $\int Y_{lm}^* \hat{l}_x Y_{lm} \hat{l}_x = \int Y_{lm}^* \frac{1}{i\hbar} [\hat{l}_y, \hat{l}_z] Y_{lm} = \frac{1}{i\hbar} \int Y_{lm}^* \hat{l}_y \hat{l}_z Y_{lm} - Y_{lm}^* \hat{l}_z \hat{l}_y Y_{lm} = 0$ so we get :

$$\overline{\hat{l}_x} = \overline{\hat{l}_y} = 0 \quad (45)$$

so we know the uncertainty relation between $\hat{l}_x$ and $\hat{l}_y$

$$\Delta \hat{l}_x \Delta \hat{l}_y = \sqrt{\overline{\hat{l}_x^2} - \overline{\hat{l}_x}^2} \sqrt{\overline{\hat{l}_y^2} - \overline{\hat{l}_y}^2} = \frac{1}{2}(l(l+1) - m^2)\hbar \quad (46)$$

4.5 Function of operator

We can define the function of operators as :

$$F(\hat{A}) = \sum_n \frac{F^n(0)}{n!} \hat{A}^n \quad (47)$$

such as $e^{a \frac{d}{dx}}$ is a typically function of operator. And the effect of this function is :

$$e^{a \frac{d}{dx}} \psi(x) = \sum_n \frac{a^n}{n!} \frac{d^n}{dx^n} \psi(x) = \psi(x + a) \quad (48)$$

so $e^{a \frac{d}{dx}}$ is the operator that move the function point from x to x+a.

5 symmetry of evolving Dynamical variables

5.1 Conserved observable

If a average value of observable is not changed with time , thus this observable is a conserved observable.

$$\frac{d}{dt} \bar{A}(t) = \frac{1}{i\hbar} \overline{[A, H]} \quad (49)$$

So if the operator of this observable is commuted with Hamilton operator , this observable is a conserved observable. **Notice the detail of definition described : AVERAGE VALUE is not go with time, not the classic type "value not go with time". The reason is the principle in QM is very different with classic mechanic.** And of course a Conserved Observable can be a not certain value, so the state of the operator observed can be the non-eigenstate of this operator. The key word is "average value".

5.2 Relationship Between Conserved Observable And Energy Level Degenerate

There are three Theorems:

(A) Set two non-commuted conserved observable F and G in a system, that means: $[F, H] = 0$; $[G, H] = 0$ but $[F, G] \neq 0$, thus the energy level usually degenerated.

(B) Inference from (A): If a conserved observable F in a system and there is a level of energy not degenerated, then ψ_E is the eigenstate of F.

(C) virial theorem:

$$2\bar{T} = \overline{\mathbf{r} \cdot \nabla V} \quad (50)$$

5.3 * Schrödinger picture and Heisenberg picture

The average value derivative with t is $\frac{d}{dt} \bar{A}(t) = \frac{1}{i\hbar} \overline{[A, H]}$, now we rewrite $\psi(r, t)$ as $\hat{U}(t, 0)\psi(r, 0)$ and the $\hat{U}(t, 0)$ here is the time shift operator and $\hat{U}(0, 0) =$

1. Now we put it back into Shrödinger equation:

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \psi(r, t) &= \hat{H} \psi(r, t) \\ i\hbar \frac{\partial}{\partial t} \hat{U}(t, 0) \psi(r) &= \hat{H} \hat{U}(t, 0) \psi(r) \\ i\hbar \frac{\partial}{\partial t} \hat{U}(t, 0) &= \hat{H} \hat{U}(t, 0) \end{aligned}$$

so we get (if $\hat{H}$ is not go with time) :

$$\hat{U}(t, 0) = e^{\frac{-i\hat{H}t}{\hbar}} \quad (51)$$

or ($\hat{H}$ go with time):

$$\hat{U}(t, 0) = e^{\frac{-i \int \hat{H} dt}{\hbar}} \quad (52)$$

and we can proof:

$$\begin{aligned} \int \phi(r, t)^* \psi(r, t) &= \int \hat{U}(t, 0)^* \phi(r, 0)^* \hat{U}(t, 0) \psi(r, 0) \\ &= \int \phi(r, 0)^* \phi(r, 0) = 1 \end{aligned} \quad (53)$$

For a more common situatioin we can say the **wave function is not go woth time but operators are go with time**, and we rewrite $\phi(r, t)$ into $\hat{U}(t, 0)\psi(0)$. And the operator can be written in:

$$\begin{aligned} \overline{\hat{A}(t)} &= \int \hat{U}(t, 0)^* \psi(0)^* \hat{A} \hat{U}(t, 0) \psi(0) \\ &= \int \psi(0)^* \hat{U}(t, 0)^\dagger \hat{A} \hat{U}(t, 0) \psi(0) \\ &= \int \psi(0)^* \hat{A}(t) \psi(0) \end{aligned}$$

so we get the operator $\hat{A}(t)$ defined as

$$\hat{A}(t) = \hat{U}(t, 0)^\dagger \hat{A} \hat{U}(t, 0) \quad (54)$$

So the evolution equation of operators evolutino with time :

$$\begin{aligned} \frac{d}{dt} \hat{A}(t) &= \frac{d}{dt} (\hat{U}(t, 0)^\dagger \hat{A} \hat{U}(t, 0)) = \frac{d\hat{U}(t, 0)^\dagger}{dt} \hat{A} \hat{U}(t, 0) + \hat{U}(t, 0)^\dagger \hat{A} \frac{d\hat{U}(t, 0)}{dt} \\ &= \frac{1}{-i\hbar} \hat{U}(t, 0)^\dagger \hat{H} \hat{A} \hat{U}(t, 0) + \frac{1}{i\hbar} \hat{U}(t, 0)^\dagger \hat{A} \hat{H} \hat{U}(t, 0) = \frac{1}{i\hbar} \hat{U}(t, 0)^\dagger [\hat{A}, \hat{H}] \hat{U}(t, 0) \end{aligned}$$

Because $\hat{U}(t, 0)^\dagger \hat{H} \hat{U}(t, 0) = \hat{H}$ (easily prooved with: $\hat{A}$ and function of $\hat{A}$ are commutated) so :

$$\frac{d}{dt} \hat{A}(t) = \frac{1}{i\hbar} [\hat{A}(t), \hat{H}] \quad (55)$$

Noticed that this is quite different from eq(49) this equation is not about average value but the equation of operator. And this kind of theory is called **Heisenberg Represent**. The equation is called Heisenberg equation. As the evolution equation of operators.

5.4 The Relationship Between Conserved Observable and Symmetry

A.E.Noether: "every symmetry in nature is implicated some conserved law, the other side every conserved law is based on one of symmetry of nature"

(A) symmetry under transformation:

$$[\hat{Q}, \hat{H}] = 0 \quad (56)$$

(B) symmetry with conserved observable:

$$[\hat{F}, \hat{H}] = 0 \quad (57)$$

(C) symmetry of space translation:

$$[p, \hat{H}] = 0 \quad (58)$$

equally with Momentum Conserved.

(D) symmetry of space turning translation:

$$[\vec{l}, \hat{H}] = 0 \quad (59)$$

equally to angular momentum conserved.

(E) Some other symmetry in QM:

(5) 时间均匀性与能量守恒

(6) 空间反射对称性与宇称守恒

(7) 同位旋空间旋转对称性与同位旋守恒

Figure 1: picture from teacher's ppt

5.5 *Commutation

Here is an example.

6 Identical Particle System and The Wavefunction

6.1 identical particle and exchange operator

An identical particle is the particle that have all the same intrinsic properties with other particle. Two identical particles are indistinguishable in QM. Define an exchange operator:

$$P_{ij}\psi(q_1 \dots q_i \dots q_j \dots) = \psi(q_1 \dots q_j \dots q_i \dots) \quad (60)$$

For any observable in a system with identical particles, the observables have symmetry of exchange

$$P_{ij}F(q_1 \dots q_i \dots q_j \dots) = F(q_1 \dots q_j \dots q_i \dots) \quad (61)$$

$$[p_{ij}, H] = 0 \quad (62)$$

For a system of identical particles, we set if an exchange of identical particles happens, then the wavefunction of this system will only differ by times a constant.

$$P_{ij}\psi = C\psi \quad (63)$$

$$P_{ij}P_{ij}\psi = C^2\psi = \psi$$

$$C = \pm 1 \quad (64)$$

So that means only two possibilities will happen: the wavefunction is symmetric: $\psi = P_{ij}\psi$ or the wavefunction is anti-symmetric: $-\psi = P_{ij}\psi$.

6.2 properties of Boson and Fermion

Boson is the particle with spin is integer times of $\hbar$. And Fermion is the particle that the spin is half-integer times of $\hbar$. But a very important nature of Boson and Fermion is Identical Boson's exchange always symmetry for a system with Boson identical particles, and identical Fermion's exchange in a system with identical Fermion always anti-symmetric.

For Boson:

if $k_1 \neq k_2$

$$\psi_{k_1 k_2}^S(q_1, q_2) = \frac{1}{\sqrt{2}}(1 + P_{12})\phi_{k_1}(q_1)\phi_{k_2}(q_2) \quad (65)$$

if $k_1 = k_2$

$$\psi_{k_1 k_2}^S(q_1, q_2) = \phi_{k_1}(q_1)\phi_{k_2}(q_2) \quad (66)$$

For Fermion:

$$\psi_{k_1 k_2}^A(q_1, q_2) = \frac{1}{\sqrt{2}}(1 - P_{12})\phi_{k_1}(q_1)\phi_{k_2}(q_2) \quad (67)$$

and we can see the state of Fermion at $k_1 = k_2$ is not exist. **This is the Pauli principle.**

7 Matrix